

Math 534H - Spring 2026 - Lecture 21

①

Now let us solve the IBVP for wave or heat equation on $[0, L]$ but with ZERO NEUMANN boundary conditions.

Wave Equation
EQ

BC

IC

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & 0 < x < L, t > 0 \\ u_x(0, t) = 0 = u_x(L, t), \forall t > 0 \\ u(x, 0) = \phi(x), u_t(x, 0) = \psi(x) \end{cases}$$

As before use separation of variables, and look for solutions of the form:

$$u(x, t) = X(x) \cdot T(t) \quad \left(\begin{array}{l} X(x) \neq 0 \\ T(t) \neq 0 \end{array} \right)^{(2)}$$

Just as before we get two eigenvalue problems

$$\begin{array}{l} \text{2nd order ODE} \\ \text{2nd order ODE} \end{array} \left\{ \begin{array}{l} X''(x) = -\lambda X(x) \\ T''(t) = -\lambda c^2 T(t) \end{array} \right. \quad \underline{\lambda \text{ constant}}$$

• Let's assume as before and assume $\lambda > 0$.

Write $\lambda = \beta^2$, $\beta > 0$ ($\beta = \sqrt{\lambda}$) just as

$$\text{before} \left\{ \begin{array}{l} X(x) = C \cos(\beta x) + D \sin(\beta x) \\ T(t) = A \cos(\beta \cdot c t) + B \sin(\beta \cdot c t) \end{array} \right.$$

for A, B, C, D constants.

zero Neumann BC. tell us that

$$X'(0) = 0 = X'(L).$$

$$X'(x) = -C\beta \sin(\beta x) + D\beta \cdot \cos(\beta x).$$

$$0 = X'(0) = D\beta ; \beta > 0 \Rightarrow \boxed{D = 0}$$

$$0 = X'(L) = -C\beta \sin(\beta L) ; \beta > 0$$

$C \neq 0$ (otherwise $X(x) \equiv 0 \Rightarrow u(x, t) \equiv 0$ trivial solution)

So we must have $\sin(\beta L) = 0$

$$\Rightarrow \boxed{\beta = \frac{n\pi}{L} ; n \geq 1}$$

All in all $X(x) = C \cos\left(\frac{n\pi}{L} x\right); n \geq 1$ ④

WAIT A SECOND!

Why?

Now : $n=0$ is included because when $n=0$ we get simply that $X(x) = C$ (nontrivial constant)

In other words $n=0$ is allowed (when seeking nontrivial solutions) which implies that $\lambda = 0$

leads in the zero Neumann BC case to nontrivial solutions. So now $\lambda \geq 0$

and $n = 0, 1, 2, 3, \dots$

Now for $T''(t) = -\lambda c^2 T(t)$ we get (5)
the same as before for $\lambda > 0$. But now we
need to also find $T(t)$ when $\lambda = 0$.

We reinspect the second order eq. for T :

$$T''(t) = 0 \quad \text{when } \lambda = 0 \text{ so in this}$$

$$\text{Case } T(t) = \tilde{A}_0 + \tilde{B}_0 t.$$

While for $\lambda > 0$ we get the same as for zero

Dirichlet.

(6)

To summarize:

$$\lambda = 0 = \kappa \quad \text{we get} \quad u(x,t) = X(x)T(t) \\ = C \cdot (\tilde{A}_0 + \tilde{B}_0 t)$$

$$\text{So } \boxed{\kappa = 0} \quad u(x,t) = \frac{A_0}{Z} + \frac{B_0}{Z} t$$

which is trivial now; and

$$\frac{A_0}{Z} := C \tilde{A}_0$$

$$\frac{B_0}{Z} := C \tilde{B}_0$$

$\boxed{\kappa \neq 0}$ we get that

$$u_n(x,t) = \left[A_n \cos\left(\frac{n\pi}{L} \cdot c t\right) + B_n \sin\left(\frac{n\pi}{L} \cdot c t\right) \right] \cos\left(\frac{n\pi}{L} x\right)$$

Since the sum of solutions is also a solution we have

$$u_N(x, t) = \frac{A_0}{2} + \frac{B_0}{2}t + \sum_{n=1}^N \left[A_n \cos\left(\frac{n\pi}{L} \cdot c \cdot t\right) + B_n \sin\left(\frac{n\pi}{L} \cdot c \cdot t\right) \right] \cos\left(\frac{n\pi}{L}x\right)$$

for any $N \geq 1$.



We still have to solve - as before - for the initial conditions. So now the question is if and when

$$u_N(x, 0) = \frac{A_0}{2} + \sum_{n=1}^N A_n \cdot \cos\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \phi(x) -$$

$$\left(u_N\right)_t(x, 0) = \frac{B_0}{2} + \sum_{n=1}^N \underbrace{B_n \frac{n\pi}{L} \cdot c}_{\text{}} \cdot \cos\left(\frac{n\pi}{L}x\right) \stackrel{?}{=} \psi(x)$$

The answer to these questions (?) is going to be given by Fourier analysis / Fourier series which

⑧

will ensure that for given ϕ, ψ continuous

function we can express them as (possibly infinite)
 $N \rightarrow \infty$

Cosine sums (cosine series).

EXAMPLE:

As before if we are given special ϕ, ψ we might be able to solve directly: indeed suppose $L = \pi$

$$\phi = 3 \cos x ; \psi(x) = 1 - \cos(4x) \text{ and}$$

W-BIVP $\left\{ \begin{array}{l} \frac{c=2}{u_{tt} - 4 u_{xx} = 0, \quad 0 < x < \pi, \quad t > 0 \\ u_x(0, t) = 0 = u_x(\pi, t), \quad t > 0 \\ u(x, 0) = 3 \cos x ; \quad u_t(x, 0) = 1 - \cos(4x) \end{array} \right.$

We know that then the general solution to ⑨

$$u_{tt} - 4u_{xx} = 0 \quad \text{with} \quad u_x(0, t) = 0 = u_x(\pi, t) \quad \forall t$$

has the form * in page ⑦ which in this case
reads as (for any large $N \geq 1$) : $(c=2 \quad L=\pi)$

$$u_N(x, t) = \frac{A_0}{2} + \frac{B_0}{2}t + \sum_{n=1}^N \left(A_n \cos(2nt) + B_n \sin(2nt) \right) \cos(nx)$$

Now let's try to also satisfy the given initial conditions : we need to find suitable constants

A_0, B_0, A_n, B_n such that :

$$\phi(x) = 3 \cos x = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(nx) \quad (†) \quad (10)$$

$$\psi(x) = 1 - \cos(4x) = \frac{B_0}{2} + \sum_{n=1}^{\infty} B_n \cdot 2n \cos(nx) \quad (††)$$

IN THIS PARTICULAR CASE WE CAN GUESS WHAT THEY ARE :

Indeed : For $\phi(x)$ take :

$$A_0 = 0, \quad A_1 = 3, \quad \text{and} \quad A_n = 0 \text{ for } n \geq 2.$$

For $\psi(x)$ take : $B_0 = 2, B_1 = 0, B_2 = 0, B_3 = 0$

$$B_4 = -\frac{1}{8} \text{ and } B_n = 0 \text{ for } n \geq 5.$$

With these choices (†) and (††) hold ! and the solution to our W-BIVP Example (page 8) is :

$$u(x, t) = t + 3 \cos(2t) \cos(x) - \frac{1}{8} \sin(8t) \cos(4x)$$